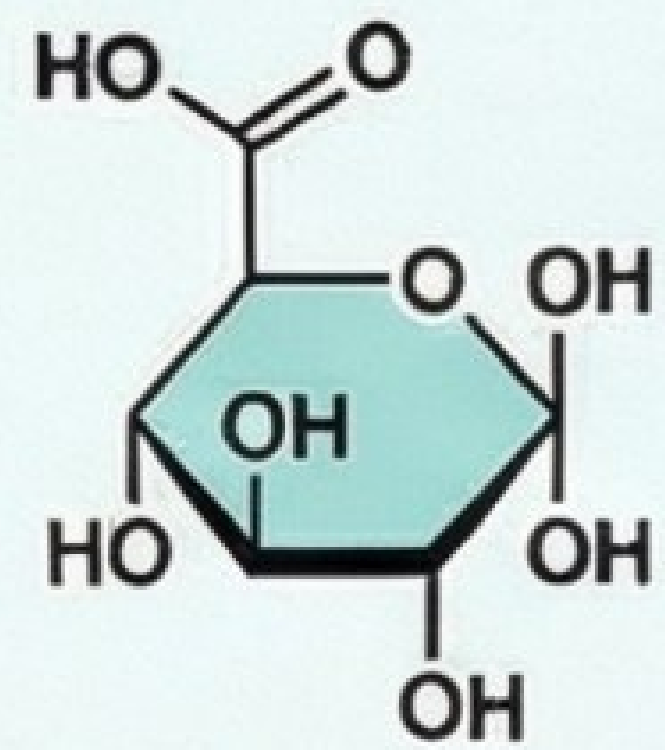


Biomolecules Unpacked: A Visual Guide to Carbohydrates & Proteins

CARBOHYDRATES

1. What are Carbohydrates?

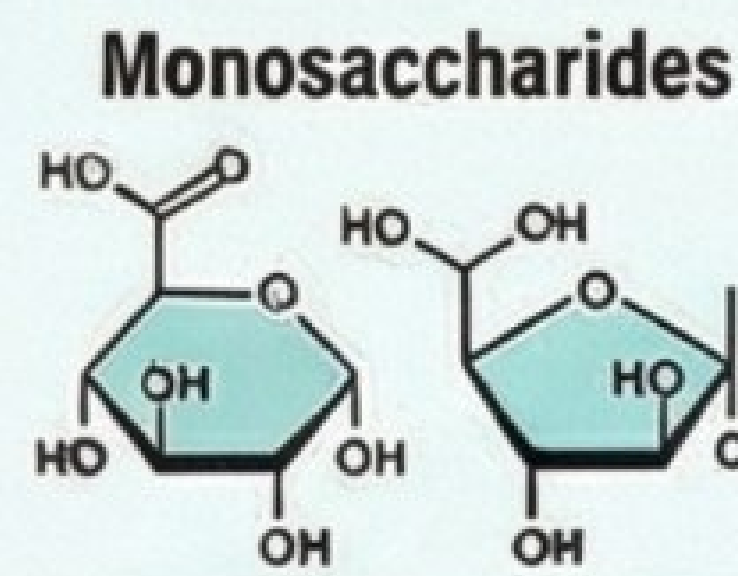


The 3 Core Conditions

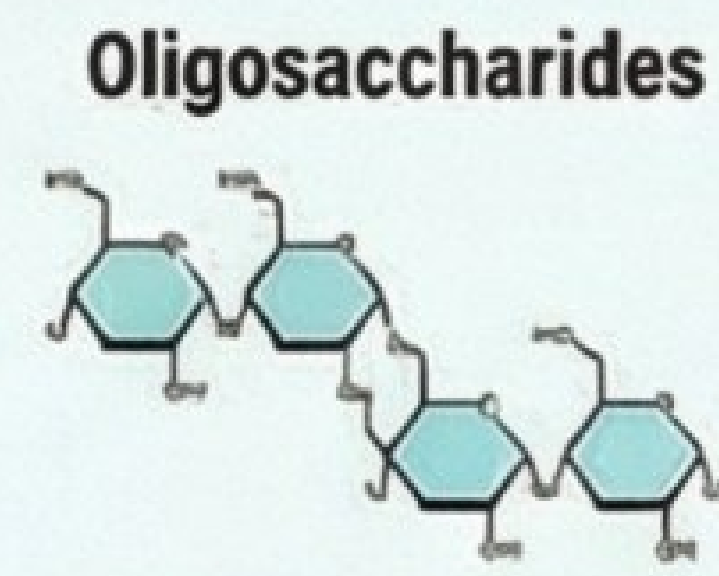
The Chemical Definition
Optically active polyhydroxy aldehydes or ketones.



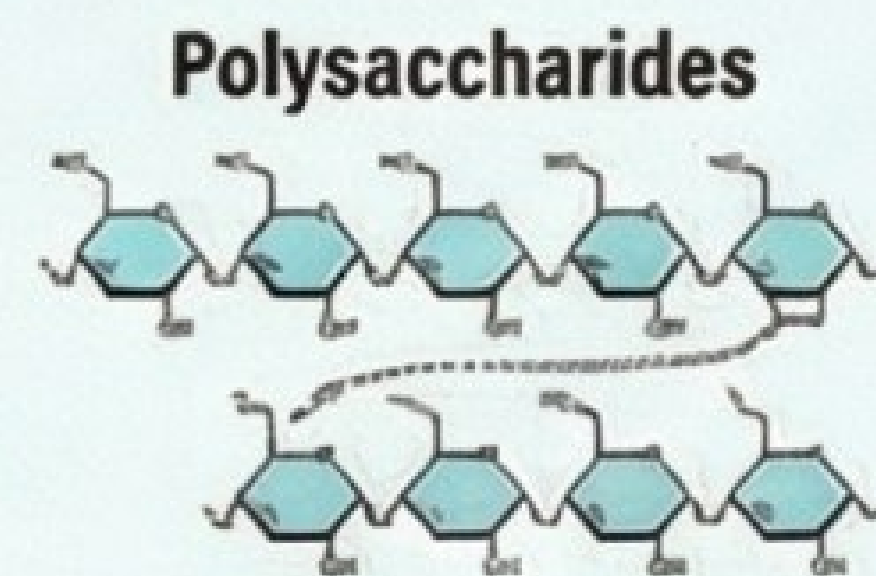
2. How Carbohydrates are Classified



Glucose - Fructose
Simplest units, cannot be hydrolyzed. Single units.



Oligosaccharides
Hold 2 to 10 monosaccharide units on hydrolysis.



Starch, Cellulose
Yield a large number of monosaccharide units. Complex polymers.

3. Reducing vs. Non-Reducing Sugars

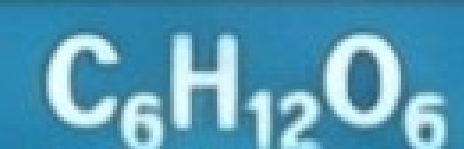


Have free aldehyde/ketone group, Reduce Tollens' or Fehling's reagents. Examples: Glucose, Fructose.



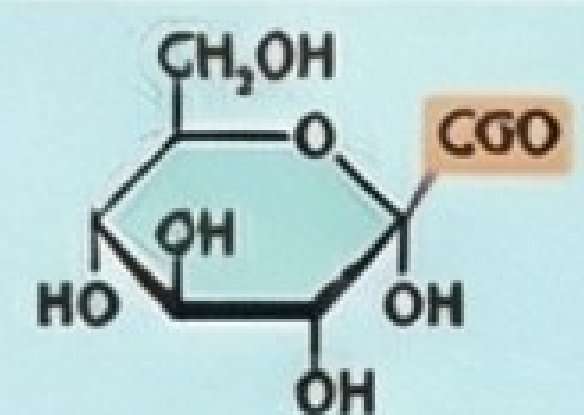
No free aldehyde/ketone group. Example: Sucrose.

4. A Closer Look: Glucose & Fructose



Identical Formula

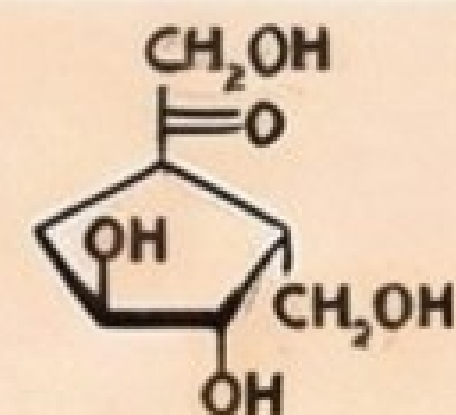
The Key Structural Difference



Aldohexose



Glucose: "Grape sugar"

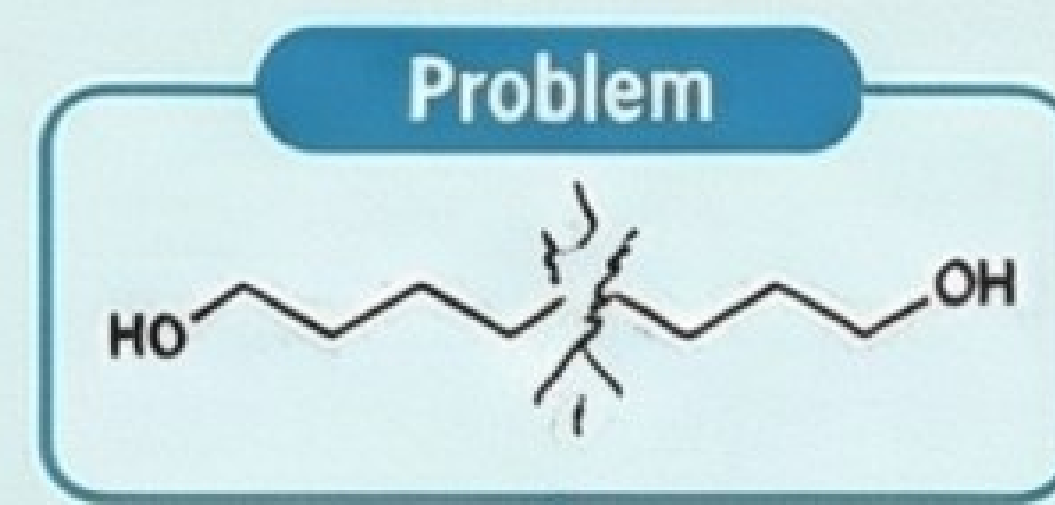


Ketohexose

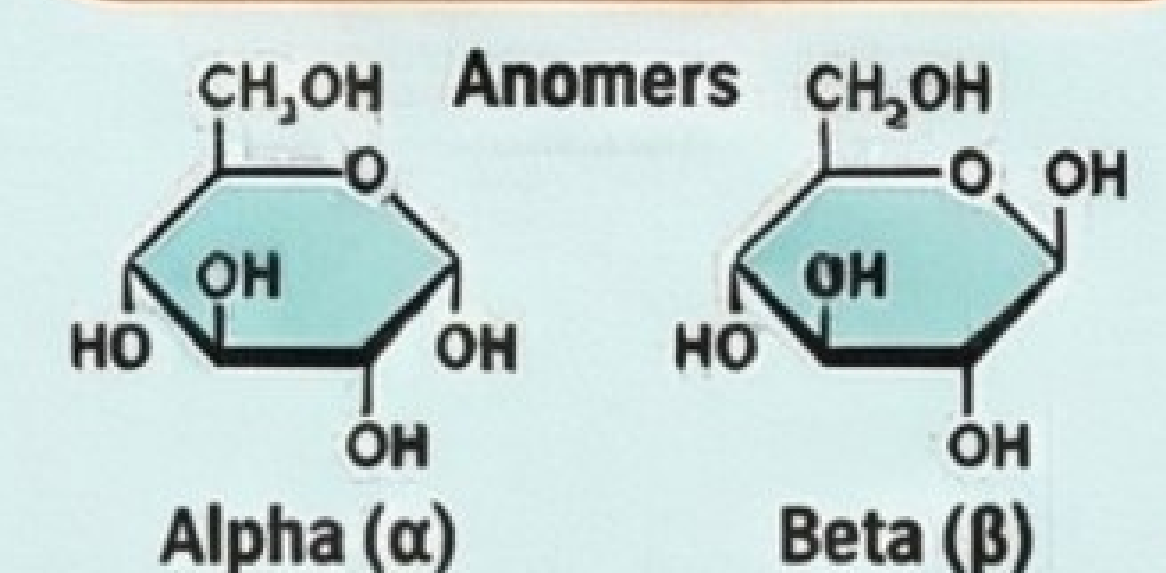
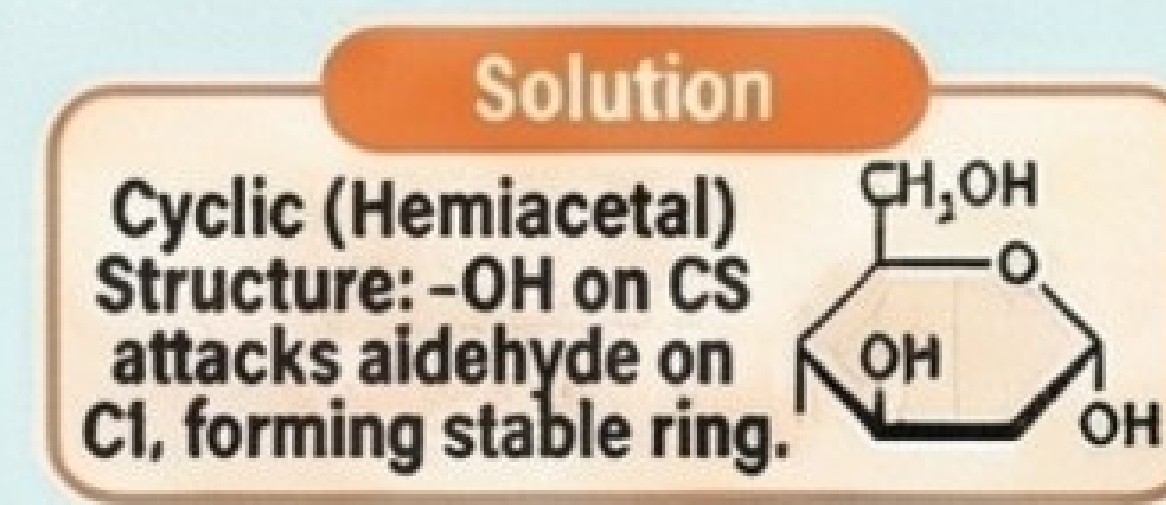


Fructose: Fruits, Honey

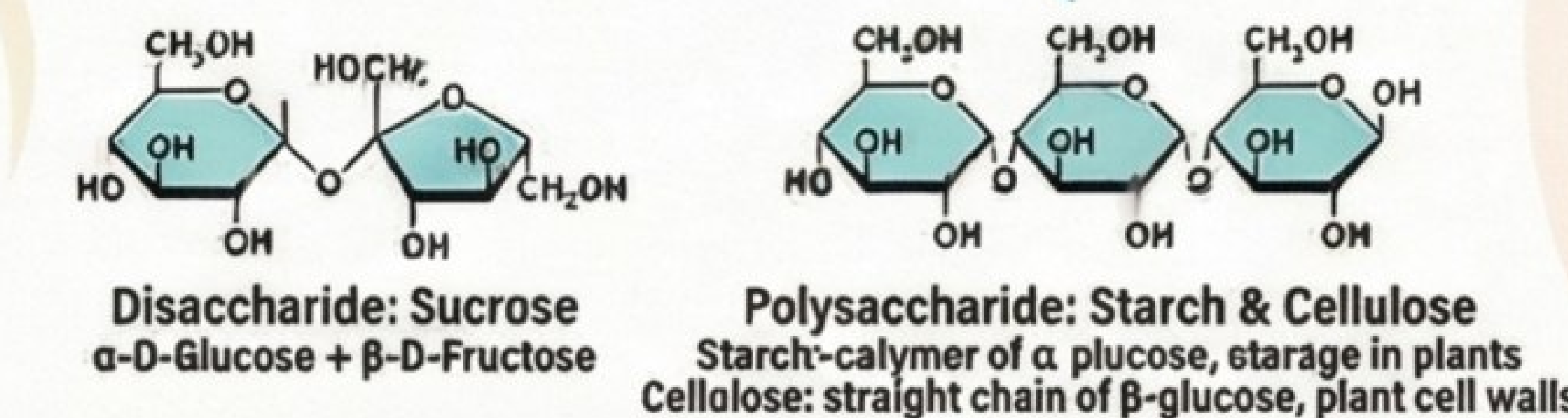
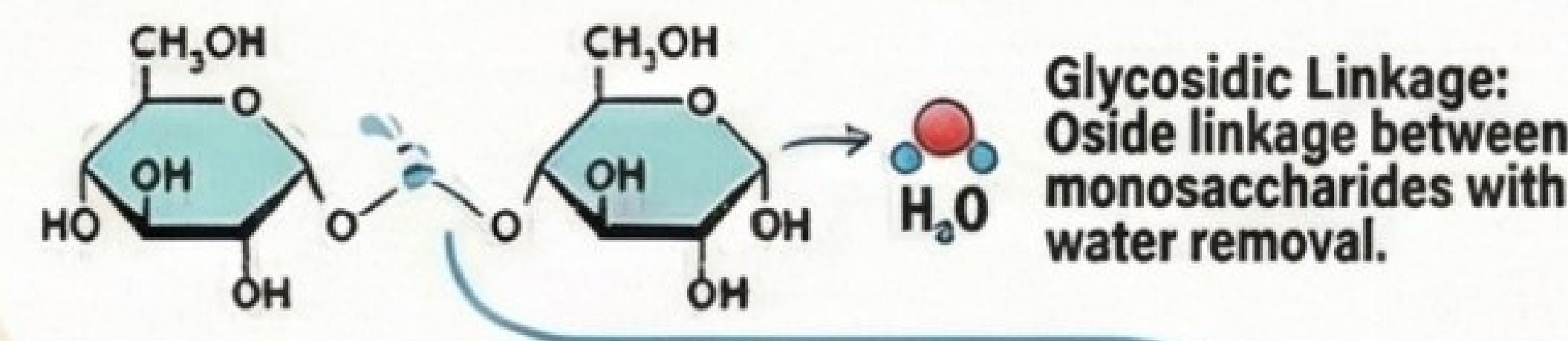
6. The Failure of the Linear Structure & Rise of the Cyclic Form



These two cyclic forms, which differ only in the configuration of the -OH group on the first carbon (C1), are called **anomers**.

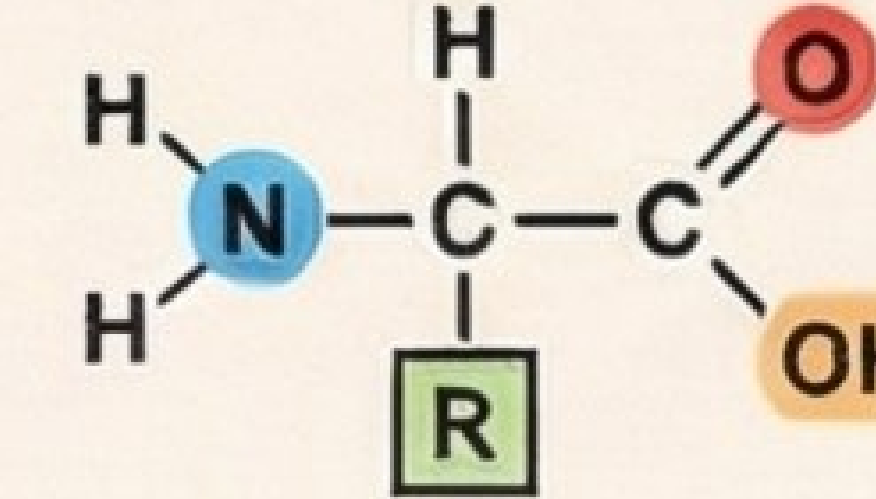


7. Building Larger Carbohydrates

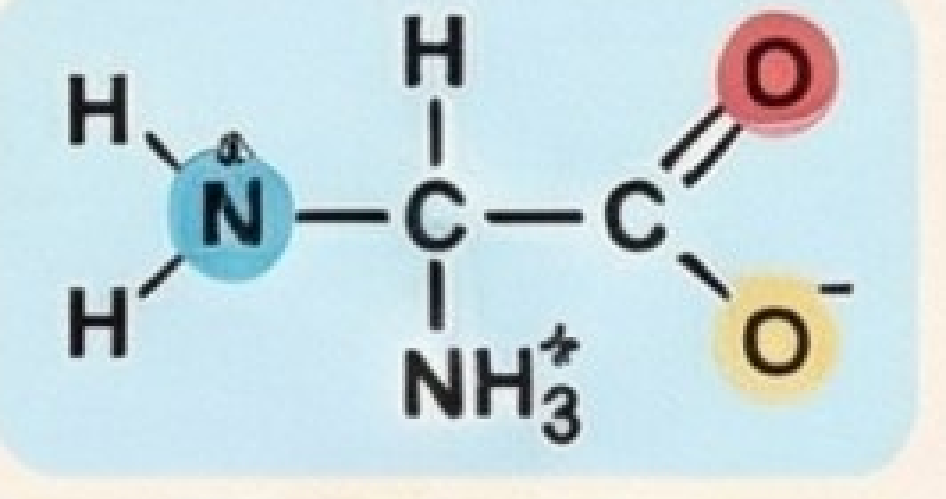


PROTEINS

8. Amino Acids: The Building Blocks



Molecules containing both basic amino and acidic carboxyl groups.



The Zwitterion
In aqueous solution, forms dipolar ion. High melting points.

Essential vs. Non-Essential



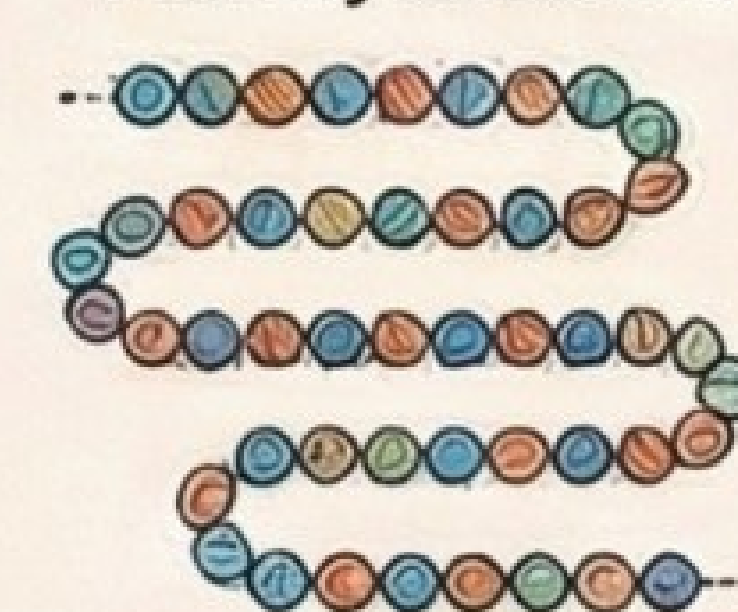
Essential
e.g., Valine
Cannot be synthesized by body. Must be obtained from diet.



Non-Essential
e.g., Glycine
Can be made by the body.

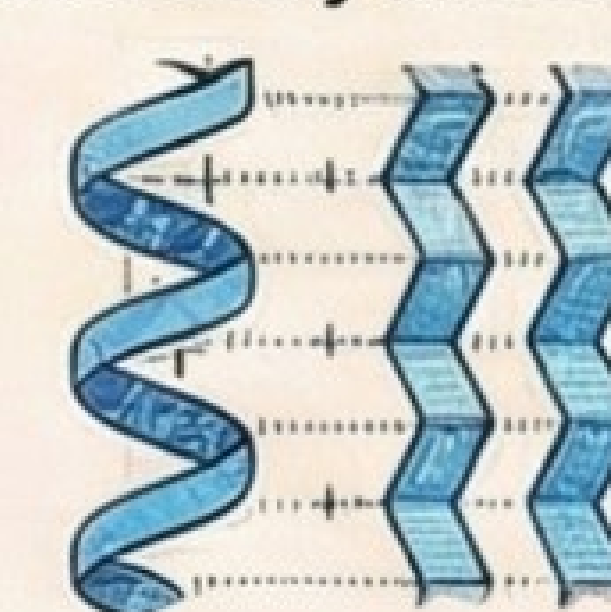
9. The Four Levels of Protein Structure

Primary Structure



Specific sequence of amino acids in polypeptide chain.

Secondary Structure



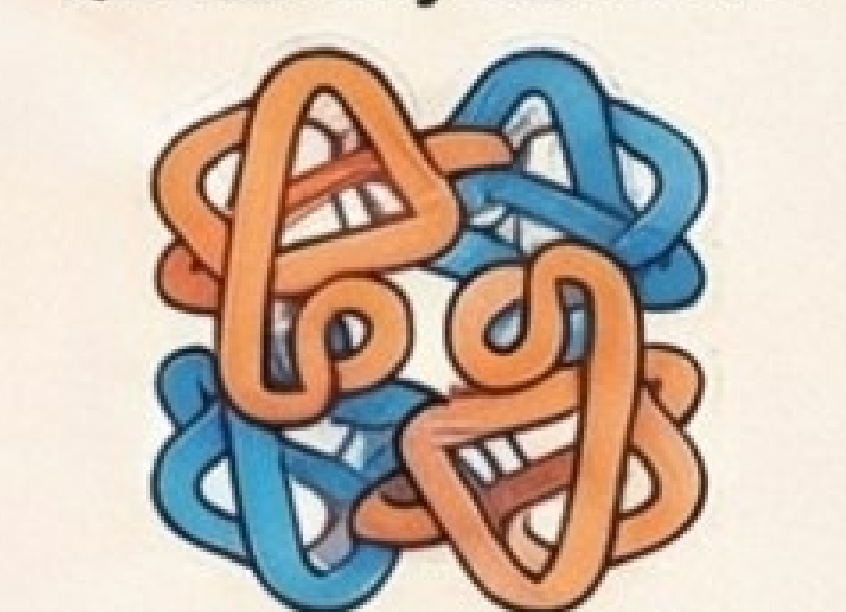
Local shape: α -Helix or β -Pleated Sheet, stabilized by hydrogen bonds.

Tertiary Structure



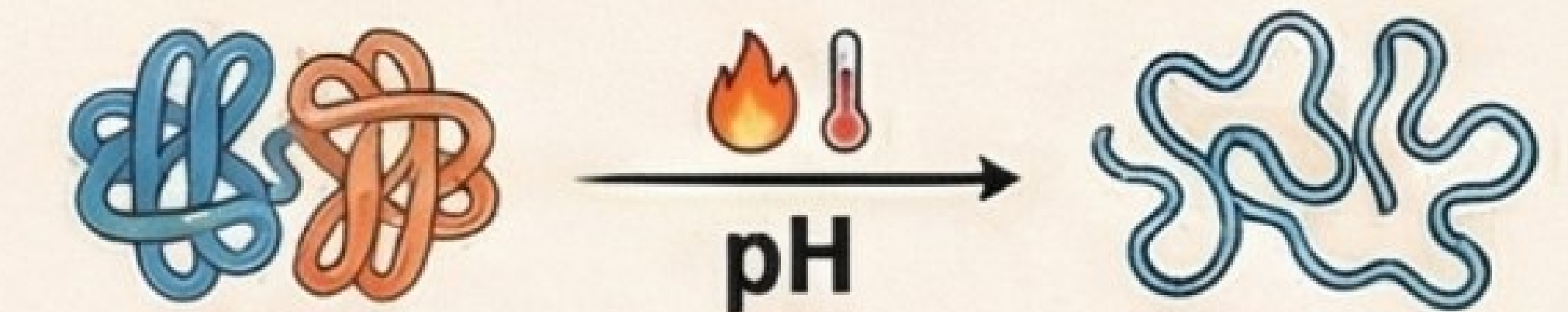
Overall 3D folding of polypeptide chain.

Quaternary Structure



Arrangement of multiple polypeptide chains (subunits).

10. Denaturation: When Proteins Unravel



Before
Process where protein loses biological activity due to disruption of secondary & tertiary structures.



Caused by heat or pH changes. Primary structure remains intact.

Real-World Example: Coagulation of egg white when boiled.